UU-T-595C
July 27, 1976
SUPERSEDING
Fed. Spec. UU-T-595B
April 4, 1967

FEDERAL SPECIFICATION

TOWEL, WIPING, PAPER: INDUSTRIAL AND INSTITUTIONAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers disposable paper towels for general purpose wiping, cleaning, and polishing.

1.2 Classification. Towels covered by this specification shall be of the following types as specified (see 6.2).

Type	Basis weight (24 x 36-500)
I	11 pounds
II	18 pounds
IV	34 pounds
V	45 pounds
VI	40 pounds (reinforced)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets, and Related Products.
T 404 - Tensile Breaking Strength of Paper and Paperboard.
T 410 - Weight Per Unit Area (Basic Weight or Substance) of Paper and Paperboard.
T 432 - Water Absorption of Bibulous Paper.
T 456 - Wet Tensile Breaking Strength of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Paper. Paper towels shall be made of fibrous cellulosic material which may include reclaimed fibers.

3.1.2 Reinforcing material. Reinforcing material for type VI towels shall be yarns of either natural or synthetic origin so laid and bonded that the resultant towels conform to the physical characteristics as specified in table II.

3.2 Construction. Towels covered by this specification may be single or multi-ply construction. The plies of multi-ply wiping towels shall be securely joined by adhesive bonding or by mechanical means. Type VI wiping towels shall contain a reinforcing material between the plies.

3.2.1 Color and surface finish. Wiping towels may be bleached or natural (unbleached) color at the suppliers' option. The surface shall be creped or embossed at the option of the supplier.

3.2.2 Form. Towels shall be furnished in flat sheets or folded, in accordance with the suppliers' commercial practice.

3.2.3 Size and area. Towels for the size and type specified shall meet the smaller dimension and minimum area requirements indicated in table I.

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TABLE I. Sizes and surface areas

Type	Smaller dimension (min) (inches)	Area (min) (square inches)
I	4-7/8	40
	8-1/8	125
	14-1/2	250
II	11-5/8	200
IV	9-1/4	125
V	12-5/8	175
VI	12-5/8	200
	12-5/8	280
VI (Roll)	10-1/2	230 (per perforated sheet)

3.3 Physical requirements. Towels shall conform to the physical characteristics in table II when tested in accordance with 4.5.

TABLE II. Physical properties of paper towels

Properties	Types I	II	IV	V	VI
Basis weight, lbs. (min) (24 x 36-500)	11	18	34	45	40
Tensile strength, oz/in. (min)					
Dry					
Machine direction	15	30	70	40	65
Cross direction	7	15	30	25	35
Wet					
Machine direction	5	9	20	12	35
Cross direction	2	3	9	8	25
Absorptiveness rate seconds (max)					
0.1 ml water	10*	10*	14	6	8
0.1 ml oil	15*	15*	50	12	-
Total absorption Percent of dry wt. (min)					
Water	275	300	300	500	500
Oil	150	150	150	300	225

*For single ply towels, the specimen shall have two thicknesses of the wiper, placed one upon another.

3.4 Workmanship. Wiping towels shall be free of holes, cuts, tears, and wrinkles. The edges shall be straight or serrated, cleanly cut, and shall not be ragged. Surfaces of the wiping towels shall be clean; there shall be no evidence of foreign materials or dirt. The plies of multi-ply towels shall be securely joined. Wipers shall be free of defects that affect serviceability or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated. For sampling purposes a lot shall consist of all wiping towels of the same type and size offered for delivery at one time.

4.2 Inspection levels. The inspection levels for establishing sample size and the acceptable quality levels (AQL's) expressed as percent defective shall be as follows:

Sample unit	Inspection level	AQL
Shipping container	S-2	4.0
Dimensional defects	S-2	4.0
Visual defects	1	6.5

4.3 Examination of the preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking conform with the requirements of section 5 of this specification. The sample unit shall be one (1) shipping container, fully packed and selected prior to the closing operation. Closed shipping containers shall be examined for closure defects and workmanship.

Examine	Defect
Marking (interior and exterior)	Incorrect; incomplete; illegible; omitted. Improper size, location, sequence, or method of application.
Materials	Any non-conforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components. Packages improperly closed, bagged, or wrapped. Bulged or distorted container. Torn or cut container.
Contents	Less than specified packages in shipping container.

4.4 Examination of the end item. The sample unit shall be one package of paper towels. Five specimens shall be selected at random from each sample unit and examined for the defects listed below. If the average of the dimensions of the specimens in any sample unit do not meet the values specified, the unit shall be considered defective.

Examine	Defect
Dimension	Size and area not as specified.
Workmanship (visual defects)	Edges not straight or serrated. Edges not fully formed; ragged. Any cuts, tears, holes, or wrinkles. Surface not clean. Any stains, dirt, or foreign material. Plies not securely joined.

4.5 Testing of the end item. Each lot of wiping towels of one type and size shall be tested for the characteristics in table III. The sample unit shall be ten (10) wiping towels, full size, randomly drawn from a package. No more than one (1) sample unit shall be drawn from any one shipping container. Five sample units shall be tested. The average value for each characteristic shall meet the requirements specified.

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TABLE III. Instructions for testing and item

Characteristics	Specification requirement paragraph	Test method	Requirement applicable to	Number determinations per sample unit	Results reported to nearest
Atmospheric conditions		T-402	All testing		
Basis weight	Table I	T-410	Lot average	10	0.1 pound
Tensile strength					
Dry	Table I	T-404	Lot average	10 each direction	0.1 oz./in
Wet	Table I	T-456	Lot average	10 each direction	0.1 oz./in
Absorptiveness rate					
Water	Table I	T-432	Lot average	5	second
Oil	Table I	4.7	Lot average	5	second
Total absorption					
Water	Table I	4.6	Lot average	5	1 percent
Oil	Table I	4.6	Lot average	5	1 percent

4.6 Total absorption.

4.6.1 Water. Samples are conditioned in accordance with TAPPI T-402. Size 4 x 4 ± 1/8 inch specimens are weigh and immersed into 25°C. water for 3 minutes (± 10 seconds). The specimens are then removed from the water by lifting them at one corner until their surfaces are vertical to the floor, thus allowing the excess water to drain off for 60 ± 10 seconds. They are then placed in tared weighing bottles and reweighed. The percent absorption is calculated as follows:

$$\% \text{ absorption} = \frac{\text{Wet weight} - \text{Dry weight}}{\text{Dry weight}} \times 100$$

4.6.2 Oil. Test in accordance with 4.6.1 and use oil in lieu of water. The drain time for the immersed samples should be 20 ± 2 minutes.

4.7 Absorptiveness rate (oil). Test in accordance with TAPPI T-432 using 0.1 ml of white mineral oil, 80-90 S.U. (Saybolt Universal) viscosity, +30 Saybolt color, N.F. grade instead of water.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Paper wiping towels of one type and size shall be packaged in the quantities indicated in table IV, in paperboard boxes conforming to PPP-B-566, or sleeve wrapped with kraft paper having a minimum basis weight of 30 pounds (24 x 36-500), as applicable. The style of box is optional with the supplier and may be a dispenser type box. When specified (see 6.2), sleeve wrapped wiping towels shall be overwrapped with plastic film not less than 0.001 inch thick. Towels in roll form shall be wrapped with kraft paper or plastic film.

5.1.1.1 Sea packaging (type V, VI, only). When specified (see 6.2), type V and VI towels, as specified, shall be formed into bundles of fifty (50) towels and sleeve wrapped with kraft paper specified in 5.1.1. Two (2) sleeve wrapped bundles shall be packaged in a snug fitting polyethylene bag made of not less than 1.5 mil material and closed with a twisted wire closure. Rolls shall be wrapped as specified in 5.1.1.

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TABLE IV. Packaging and packing

Type	Size (minimum area) square inches	Wipers per package	Type of package	Packages per shipping container
I	40	240	Box	72
	125	200	Box	24
	250	100	Box	24
II	200	100	Box	24
IV	125	125	Box	18
V	175	50	Box or Sleeved	24
VI	200	100	Sleeved	10
	280	100	Sleeved	10
(Rolls)	230	60	Roll	12

5.1.2 Level B. Level B packaging shall be the same as 5.1.1 except that no overwrap shall be required (see 6.3).

5.1.3 Level C. Paper wiping towels in the quantities listed in table IV for type and size shall be packaged to afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Paper wiping towels, packaged as specified in 5.1, shall be packed in snugfitting fiberboard containers conforming to class-weather resistant V3c or V3s, style RSC of PPP-B-636. The number of packages per shipping container shall be as indicated in table IV for each type and size. Each shipping container shall be closed, waterproofed with tape and reinforced with tape banding in accordance with the appendix to the container specification.

5.2.1.1 Sea packing (type V, VI, only). When specified (see 6.2), towels packaged as specified in 5.1.1.1, shall be packed in snug fitting fiberboard boxes conforming to style RSC, V3c or V3s or PPP-B-636. Each box shall be closed and waterproofed with tape in accordance with the appendix to the box specification. Boxes shall be designed to pass freely through a 24-inch diameter opening.

5.2.2 Level B. The number of packages of paper towels indicated in table IV for each type and size, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard container conforming to PPP-B-636, type CF or SF, class domestic, style RSC, with a minimum bursting strength of 200 p.s.i. The shipping containers shall be closed in accordance with the appendix to the container specification.

5.2.3 Level C. Paper wiping towels packaged as specified in 5.1.3 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with common carrier rules or regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking specified by the contract or order, unit and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Paper wiping towels covered by this specification are intended for use as general purpose disposable wipers for cleaning, and polishing and wiping operations on dry, wet, or oily surfaces. Wipers are not intended for use on transparent plastic surfaces.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2.1 and table II).
(c) Applicable level of packaging and packing (see 5.1 and 5.2).
(d) Whether plastic overwrap is required for level A packaging (see 5.1.1).
(e) Whether sea packaging and packing is required for level A use (see 5.1.1.1 and 5.2.1.1).

Preparing Activity:

GSA - FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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